A proper spectral modification violating Efimov rigid-fibre invariance

Theorem. *Let*

$$R = H(\mathbf{Q}[[t]]), \qquad J = \bigoplus_{n\geq 1} \Sigma^n (R/t^n), \qquad A = R \oplus J,$$

where A is the split square-zero E_∞-extension, and put $K = R[1/t] \simeq A[1/t]$. Then $\operatorname{Spec} R \to \operatorname{Spec} A$ *is proper and locally almost of finite presentation and is an equivalence over $D(t)$, but*

$$\operatorname{Rig}(\operatorname{Mod}_A^{t\text{-comp}}) \otimes_{\operatorname{Mod}_A} \operatorname{Mod}_K \longrightarrow \operatorname{Rig}(\operatorname{Mod}_R^{t\text{-comp}}) \otimes_{\operatorname{Mod}_R} \operatorname{Mod}_K \tag{1}$$

is not fully faithful.

Proof. Every bounded truncation of J is a finite sum of perfect R-modules, hence $J \in \operatorname{APerf}(R)$; the square-zero criterion makes $A \to R$ locally almost of finite presentation. Its classical truncation is the identity, hence the induced spectral morphism is proper, and $J[1/t] \simeq 0$ gives $A[1/t] \simeq R[1/t]$.

Completed scalar extension

$$\Phi(M) = \Lambda_t(R \otimes_A M) : \operatorname{Mod}_A^{t\text{-comp}} \rightleftarrows \operatorname{Mod}_R^{t\text{-comp}} : \operatorname{Res}$$

induces $F \dashv G$ on rigidifications. Let a_A, a_R be the standard maps from rigidification and Λ_A, Λ_R their fully faithful symmetric-monoidal right adjoints. We use

$$a_R F \simeq \Phi a_A, \qquad G\Lambda_R \simeq \Lambda_A \operatorname{Res},$$

and the resulting unit calculation

$$\operatorname{fib}\big(\mathbf{1} \to GF(\mathbf{1})\big) \simeq \Lambda_A(J). \tag{2}$$

After scalar extension to K, the unit fibre is

$$L_t\Lambda_A(J) := \operatorname{colim}\big(\Lambda_A(J) \xrightarrow{t} \Lambda_A(J) \xrightarrow{t} \cdots\big). \tag{3}$$

Set $Q = \bigoplus_{m\geq 1} \Sigma^{-m} R$, with A-action through $A \to R$, and put $\mathcal{D} = (\operatorname{Mod}_A^{t\text{-comp}})^{\omega_1}$. The shifts make J, Q derived t-complete and ω_1-compact. Efimov's predual description gives

$$\operatorname{Map}\big(Q^{op}, \Theta_A L_t \Lambda_A(J)\big) \simeq \operatorname{Map}_{\operatorname{Mod}_A^{t\text{-comp}}}\big(\mathbf{1}, J \widehat{\otimes}_A Q\big)[1/t], \tag{4}$$

where $\Theta_A : \operatorname{Rig}(\operatorname{Mod}_A^{t\text{-comp}}) \hookrightarrow \operatorname{Ind}(\mathcal{D}^{op})$ is the canonical fully faithful functor.

The standard algebra retract in Mod_R makes $J \otimes_R Q$ a retract of $J \otimes_A Q$. Its equal-index summands give a further retract

$$D := \bigoplus_{n\geq 1} R/t^n \;\hookrightarrow\; J \otimes_A Q, \tag{5}$$

so $\Lambda_t D$ is a retract of $J \widehat{\otimes}_A Q$. Clearly $D[1/t] \simeq 0$. For $r \geq 1$, the compatible elements

$$x^{(r)} = \big(t^{\lfloor n/2 \rfloor} \bmod t^{\min(n,r)}\big)_{n\geq 1} \in \pi_0(D \otimes_R^{\mathbf{L}} R/t^r)$$

have finite support for fixed r and define an inverse-limit class killed by no power of t. The Milnor exact sequence therefore gives

$$(\Lambda_t D)[1/t] \not\simeq 0.$$

By (4)–(5), $L_t\Lambda_A(J) \not\simeq 0$. Hence the localized adjunction unit is not invertible, and (1) is not fully faithful. ☐

References: A. I. Efimov, *Localizing invariants of inverse limits*, arXiv:2502.04123v2, Prop. 1.27, Prop. 1.34, Thm. 4.2 and Cor. 4.4; the standard square-zero criterion for locally almost finitely presented spectral algebras.